

Year 7 • Chapter 7 • Decimals: answers (teacher copy)

7.01 Ordering decimals (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: 0.7
- Ex 1b: 34.89
- Ex 2a: 8 hundredths
- Ex 2b: 3.998, 34.89, 34.98, 36.86, 38.141, 340
- Ex 3a: 1.293, 1.239, 1.1015, 1.015, 1.006
- Ex 3b: e.g. 2.355

Easy (drill)

- | | | |
|----------|-------------------|-------------------|
| 1. 8.42 | 12. 3.5 | 23. 9 hundredths |
| 2. 1.69 | 13. 5.6 | 24. 5 ones |
| 3. 2.7 | 14. 4.71 | 25. 5 hundredths |
| 4. 8.8 | 15. 9.15 | 26. 1 thousandths |
| 5. 3.92 | 16. 3.2 | 27. 8 tenths |
| 6. 9.64 | 17. 2 thousandths | 28. 3 thousandths |
| 7. 3.39 | 18. 7 tenths | 29. 7 ones |
| 8. 8.22 | 19. 3 tenths | 30. 2 thousandths |
| 9. 6.91 | 20. 9 thousandths | 31. 4 thousandths |
| 10. 8.55 | 21. 8 thousandths | 32. 2 thousandths |
| 11. 1.9 | 22. 8 ones | |

Medium (drill)

- | | | |
|------------------------------|------------------------------|-------|
| 1. 1.34, 1.4, 1.579,
1.85 | 5. 1.68, 1.75,
1.824, 1.9 | 9. > |
| 2. 2.15, 2.685, 2.8,
2.89 | 6. 2.5, 2.53, 2.672,
2.98 | 10. < |
| 3. 3.64, 3.72, 3.8,
3.965 | 7. 3.51, 3.578, 3.81,
3.9 | 11. > |
| 4. 5.1, 5.38, 5.52,
5.96 | 8. 0.8, 0.897, 0.93,
0.96 | 12. < |
| | | 13. > |
| | | 14. > |
| | | 15. > |
| | | 16. < |

Challenging

- 1.293, 1.239, 1.1015, 1.015, 1.006
- e.g. 2.355
- 12.018 s

Extension

- 0.605
- 9 (4.31 to 4.39)

Exit ticket

- Level 1: 0.4
- Level 2: 5.02, 5.2, 5.22
- Level 3: 0.33, 0.303, 0.3, 0.033

7.02 Decimals and fractions (booklet p. 7)

WE DO examples (teacher)

- Ex 1a: $\frac{1}{2}$
- Ex 1b: $\frac{89}{100}$
- Ex 2a: $\frac{111}{250}$
- Ex 2b: 0.4
- Ex 3a: 0.07
- Ex 3b: 0.011

Easy (drill)

- | | | |
|---------------------|---------------------|-----------|
| 1. $\frac{1}{25}$ | 10. $\frac{1}{100}$ | 20. 0.1 |
| 2. $\frac{1}{50}$ | 11. $\frac{3}{10}$ | 21. 0.86 |
| 3. $\frac{33}{100}$ | 12. $\frac{17}{25}$ | 22. 0.4 |
| 4. $\frac{43}{100}$ | 13. $\frac{1}{5}$ | 23. 0.24 |
| 5. $\frac{9}{100}$ | 14. $\frac{11}{25}$ | 24. 0.89 |
| 6. $\frac{9}{10}$ | 15. $\frac{3}{50}$ | 25. 0.56 |
| 7. $\frac{4}{5}$ | 16. $\frac{4}{25}$ | 26. 0.15 |
| 8. $\frac{7}{100}$ | 17. 0.47 | 27. 0.98 |
| 9. $\frac{41}{50}$ | 18. 0.35 | 28. 0.188 |
| | 19. 0.96 | 29. 0.017 |
| | | 30. 0.404 |
| | | 31. 0.92 |
| | | 32. 0.38 |

Medium (drill)

- | | | |
|-----------------------|------------------------|------------------------|
| 1. $\frac{177}{250}$ | 9. $\frac{171}{200}$ | 17. 3 $\frac{37}{100}$ |
| 2. $\frac{927}{1000}$ | 10. $\frac{619}{1000}$ | 18. 2 $\frac{3}{4}$ |
| 3. $\frac{47}{125}$ | 11. $\frac{963}{1000}$ | 19. 3 $\frac{11}{20}$ |
| 4. $\frac{111}{200}$ | 12. $\frac{951}{1000}$ | 20. 4 $\frac{1}{4}$ |
| 5. $\frac{661}{1000}$ | 13. $\frac{157}{1000}$ | 21. 3 $\frac{33}{50}$ |
| 6. $\frac{143}{500}$ | 14. $\frac{277}{1000}$ | 22. 2 $\frac{41}{50}$ |
| 7. $\frac{487}{1000}$ | 15. $\frac{413}{500}$ | |
| 8. $\frac{409}{500}$ | 16. $\frac{759}{1000}$ | |

Challenging

- 0.6
- 0.35
- 0.036

Extension

- 0.075
- 4 $\frac{3}{40}$

Exit ticket

- Level 1: $\frac{7}{10}$
- Level 2: 0.23
- Level 3: $\frac{7}{20}$

7.03 Adding and subtracting decimals

(booklet p. 11)

WE DO examples (teacher)

- Ex 1a: 16.88
- Ex 1b: 2.379
- Ex 2a: 27.595
- Ex 2b: 104.49
- Ex 3a: 396.51
- Ex 3b: 65.355

Easy (drill)

- | | | |
|---------|----------|-----------|
| 1. 12.5 | 9. 9.3 | 17. 12.88 |
| 2. 3.4 | 10. 7.8 | 18. 8.19 |
| 3. 8 | 11. 2.2 | 19. 13.97 |
| 4. 12.2 | 12. 12.9 | 20. 1.73 |
| 5. 10.3 | 13. 10.4 | 21. 2.21 |
| 6. 0.8 | 14. 14.9 | 22. 1.91 |
| 7. 12.6 | 15. 12.4 | 23. 7.29 |
| 8. 15.9 | 16. 1.4 | 24. 9.6 |

25. 6.86	28. 1.08	31. 12.04
26. 5.53	29. 9.94	32. 6.92
27. 4.82	30. 15.48	

Medium (drill)

1. 35.32	9. 129.56	17. 4.16
2. 179.46	10. 40.01	18. 3.97
3. 34.99	11. 109.41	19. 10.99
4. 19.3	12. 369.41	20. 13.4
5. 17.4	13. 321.99	21. 3.16
6. 41.11	14. 10.16	22. 8.41
7. 18.02	15. 5.56	
8. 38.22	16. 150.1	

Challenging

- 65.355
- 396.51
- \$29.75

Extension

- 8.43
- $6.42 + 5.31 = 11.73$ (or similar)

Exit ticket

- Level 1: 4.3
- Level 2: 6.55
- Level 3: 19.08

7.04 Multiplying and dividing by powers of 10 (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: 75.4
- Ex 1b: 754
- Ex 2a: 7 540
- Ex 2b: 1.39
- Ex 3a: 0.139
- Ex 3b: 0.0139

Easy (drill)

1. 4 120	12. 9.08	23. 17.47
2. 595	13. 4.48	24. 81.92
3. 66	14. 18.3	25. 73.67
4. 6.55	15. 295	26. 734.1
5. 9 100	16. 76 400	27. 33.32
6. 92 300	17. 11.37	28. 8.182
7. 831	18. 3.605	29. 737.8
8. 932	19. 94.14	30. 1.841
9. 49 300	20. 6.96	31. 159
10. 43 100	21. 7.035	32. 0.5658
11. 2 490	22. 956.4	

Medium (drill)

1. 0.00635	9. 71 700	17. 10
2. 43	10. 8.9	18. 10
3. 0.00705	11. 63	19. 1 000
4. 963	12. 37.9	20. 100
5. 99.6	13. 0.686	21. 10
6. 9 600	14. 0.0518	22. 10
7. 27.5	15. 0.00123	23. 10
8. 0.0014	16. 5 410	24. 1 000

Challenging

- \$135
- \$8.75
- 0.096 mm

Extension

- 7 200

2. 0.16

Exit ticket

- Level 1: 38
- Level 2: 0.526
- Level 3: 49

7.05 Using a known product (booklet p. 19)

WE DO examples (teacher)

- Ex 1a: 319.2
- Ex 1b: 31.92
- Ex 2a: 319.2
- Ex 2b: 3.192
- Ex 3a: 319.2
- Ex 3b: 0.3192

Easy (drill)

1. 0.972	9. 0.056	17. 0.048
2. 0.1178	10. 0.028	18. 0.014
3. 10.12	11. 0.033	19. 0.0072
4. 8.12	12. 0.18	20. 0.9
5. 1.189	13. 0.008	21. 0.27
6. 64.4	14. 0.1	22. 0.6
7. 0.594	15. 0.081	23. 3.6
8. 0.0342	16. 1.8	24. 0.0008

Medium (drill)

1. 1.8	5. 2.2	9. 4 420
2. 0.24	6. 1.9	10. 12 250
3. 3.4	7. 4.7	11. 660
4. 3.5	8. 0.45	12. 682

Challenging

- 45
- 45
- 0.045

Extension

- 17
- 24

Exit ticket

- Level 1: 32.2
- Level 2: 3.22
- Level 3: 0.0322

7.06 Multiplying decimals (booklet p. 23)

WE DO examples (teacher)

- Ex 1a: 16.5
- Ex 1b: 0.6
- Ex 2a: 7.32
- Ex 2b: 1.3325
- Ex 3a: 0.001
- Ex 3b: 5.137

Easy (drill)

1. 26	12. 6	23. 0.08
2. 58.1	13. 40.5	24. 0.015
3. 10.8	14. 8.2	25. 0.12
4. 5.4	15. 20.7	26. 0.1
5. 65.8	16. 48.3	27. 0.002
6. 8.4	17. 0.18	28. 0.27
7. 60	18. 0.035	29. 0.007
8. 11.2	19. 0.018	30. 0.3
9. 60.8	20. 0.01	31. 0.15
10. 7.6	21. 0.06	32. 0.009
11. 25.5	22. 0.027	

1. 0.744	8. 2.408	15. 2.013
2. 1.6	9. 0.782	16. 1.587
3. 4.897	10. 2.444	17. 16.224
4. 2.352	11. 6.132	18. 8.229
5. 2.652	12. 2.94	19. 26.465
6. 1.584	13. 1.829	
7. 4.44	14. 2.47	

1. 14.4 m^2
2. \$80.33
3. 0.008

1. 0.0081
2. e.g. 0.4×0.6

1. Level 1: 8.6
2. Level 2: 0.42
3. Level 3: 3.29

1. Ex 1a: 381.2
2. Ex 1b: 12.2
3. Ex 2a: 65.2
4. Ex 2b: 0.6875
5. Ex 3a: 0.4375
6. Ex 3b: \$8.40

1. 9	12. 5.4	23. 7.16
2. 9.6	13. 8.7	24. 8.03
3. 3.8	14. 8.9	25. 9.88
4. 8.3	15. 1.1	26. 4.43
5. 5.6	16. 8.6	27. 6.89
6. 1.5	17. 8.14	28. 9.77
7. 4.9	18. 3.53	29. 1.76
8. 6	19. 1.13	30. 2.82
9. 8.7	20. 8.54	31. 3.62
10. 4.3	21. 1.47	32. 7.23
11. 7.9	22. 6.52	

1. 1	9. 1.65	17. \$6.40
2. 0.58	10. 4.3	18. \$1.48
3. 2.4	11. 0.4875	19. \$15.06
4. 0.5	12. 0.475	20. \$19.78
5. 0.3875	13. 0.625	21. \$24.85
6. 1.05	14. 1.05	22. \$20.80
7. 1.75	15. 1.4	
8. 1.2	16. 4.7	

1. \$8.40
2. 1.125 m
3. \$1.45

1. 5.6
2. 1.25 and 0.625: doubling the divisor halves the answer

1. Level 1: 3.4
2. Level 2: 2.4

1. Ex 1a: 25
2. Ex 1b: 31.8
3. Ex 2a: 230
4. Ex 2b: 25.4
5. Ex 3a: 5
6. Ex 3b: 14 cups

1. 20	12. 3	23. 4
2. 11	13. 8	24. 3
3. 24	14. 2	25. 26
4. 29	15. 15	26. 15
5. 9	16. 6	27. 19
6. 11	17. 40	28. 33
7. 8	18. 22	29. 5
8. 29	19. 21	30. 33
9. 22	20. 35	31. 39
10. 24	21. 38	32. 7
11. 5	22. 32	

1. 7.9	9. 6.7	17. 74
2. 4.4	10. 5.6	18. 79
3. 8	11. 2.5	19. 75
4. 3.4	12. 1.1	20. 5
5. 2	13. 3.9	21. 52
6. 7.8	14. 8.5	22. 76
7. 2.4	15. 4.4	
8. 6.7	16. 6.9	

1. 14 cups
2. 12.5 km/L
3. 14 m

1. 180
2. the same: both are 80

1. Level 1: 20
2. Level 2: 9
3. Level 3: 4

1. Ex 1a: 0.8
2. Ex 1b: 0.375
3. Ex 2a: 0.6
4. Ex 2b: 0.5
5. Ex 3a: 0.45
6. Ex 3b: 0.16

1. 0.125	10. 0.04	19. 0.083
2. 0.25	11. 0.6	20. 0.06
3. 0.5	12. 0.9	21. 0.3
4. 0.875	13. 0.4	22. 0.6
5. 0.55	14. 0.95	23. 1.6
6. 0.3	15. 0.85	24. 0.8
7. 0.2	16. 0.68	25. 0.1
8. 0.02	17. 0.83	26. 0.2
9. 0.625	18. 1.16	27. 0.16

28. 0.4	30. 0.7	32. 0.03	22. 3.31	23. 1.67	24. 2.50
29. 0.5	31. 1.3				

Medium (drill)

1. 2.4	10. 6.96	18. 0.23
2. 7.8	11. 4.4	19. $\frac{4}{5}$
3. 2.88	12. 5.5	20. 0.78
4. 7.48	13. 4.125	21. 0.84
5. 2.375	14. 4.5	22. 0.6
6. 4.05	15. 9.72	23. 0.53
7. 3.625	16. 6.2	24. 0.78
8. 1.625	17. $\frac{3}{5}$	
9. 3.5		

Challenging

- 0.6, $\frac{5}{8}$, $\frac{2}{3}$, 0.67
- 0.1, 0.2, 0.3, 0.7
- 0.142857 repeating: 6 digits

Extension

- $\frac{7}{9}$
- $\frac{2}{11}$

Exit ticket

- Level 1: 0.6
- Level 2: 0.875
- Level 3: 0.3

7.10 Rounding decimals (booklet p. 39)

WE DO examples (teacher)

- Ex 1a: 406.3
- Ex 1b: 125.72
- Ex 2a: 3.90
- Ex 2b: 2345.9
- Ex 3a: 78654.06
- Ex 3b: 678.431

Easy (drill)

1. 38.8	12. 87.8	23. 44.74
2. 12.5	13. 1.1	24. 69.06
3. 70.6	14. 38.3	25. 70.74
4. 17.8	15. 49.5	26. 87.02
5. 23.4	16. 70.1	27. 12.88
6. 31.8	17. 81.85	28. 83.20
7. 74.8	18. 18.00	29. 73.45
8. 86.7	19. 4.66	30. 17.70
9. 9.2	20. 25.33	31. 95.80
10. 26.0	21. 11.97	32. 35.49
11. 3.3	22. 90.53	

Medium (drill)

1. 25	8. 81	15. 89.829
2. 36.698	9. 39.989	16. 43.331
3. 31.536	10. 60.5	17. 6.75
4. 97	11. 28.539	18. 12.67
5. 89.87	12. 72	19. 5.83
6. 64.88	13. 20.4	20. 3.64
7. 100.0	14. 73	21. 2.58

Challenging

- 0.67
- \$15.74
- 4.65 and 4.74

Extension

- 23 000
- 0.0061

Exit ticket

- Level 1: 5.7
- Level 2: 12.35
- Level 3: 0.900

7.11 Decimals and money (booklet p. 43)

WE DO examples (teacher)

- Ex 1a: \$470.20
- Ex 1b: \$1.34
- Ex 2a: \$21.75
- Ex 2b: \$38.20
- Ex 3a: \$17.85
- Ex 3b: \$8.66

Easy (drill)

1. \$11.70	10. \$12.29	19. \$0.05
2. \$9.75	11. \$5.79	20. \$28.73
3. \$9.61	12. \$36.42	21. \$29.25
4. \$22.55	13. \$1.45	22. \$2.57
5. \$29.90	14. \$17.86	23. \$9.28
6. \$6.29	15. \$45.39	24. \$5.76
7. \$20.73	16. \$30.76	25. \$0.37
8. \$19.22	17. \$0.79	
9. \$22.08	18. \$25.36	

Medium (drill)

1. \$36.45	11. \$69.18	21. \$51.30
2. \$44.00	12. \$22.48	22. \$75.10
3. \$60.05	13. \$70.95	23. \$55.85
4. \$16.08	14. \$93.50	24. \$77.20
5. \$10.90	15. \$60.70	25. \$86.25
6. \$43.20	16. \$93.80	26. \$85.65
7. \$34.86	17. \$81.10	27. \$18.30
8. \$6.84	18. \$89.15	28. \$57.35
9. \$21.40	19. \$32.75	
10. \$77.77	20. \$9.90	

Challenging

- \$38.20
- \$44.60
- 2 L for \$4.60 (\$2.30/L vs \$2.40/L)

Extension

- one of each pack: 8 pens for \$10.70
- Yes: \$1.20 per pen

Exit ticket

- Level 1: \$14.60
- Level 2: \$5.65
- Level 3: \$12.75